ROI Selection for Remote Photoplethysmography

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Abstract. Camera-based remote photoplethysmography (rPPG) is a technique that can be used to measure vital signs contactlessly. In order to optimize the extraction of photoplethysmographic signals from video sequences, we investigate the spatial dependence of the photoplethysmographic signal. For an evaluation of the suitability of various regions of interest for rPPG measurements, we conducted a study on 20 healthy subjects. We analysed the videos using a refined pulse amplitude mapping approach. Our results show that the signal-to-noise ratio of rPPG signals can be improved by limiting the region of interest to certain regions of the face.

1 Introduction

Photoplethysmography (PPG) is a technique used to non-invasively determine blood volume changes at a measurement site by measuring the absorption of light. In a clinical setting PPG is used to measure oxygen saturation, but cardiac and respiratory activity can be derived from PPG signals as well. Remote photoplethysmography (rPPG) makes use of optical imaging sensors to measure subtle skin-color changes resulting from blood volume changes in near-surface vessels [1]. The contactless measurement principle makes rPPG an interesting technology in applications outside the clinical environment where traditional biomedical monitoring is not feasible today. The decreasing prices and the abundant availability of digital cameras and processing power over the last years have contributed to a growing number of rPPG related publications.

However, the image and signal-processing methods used today are not yet robust enough to allow for a reliable monitoring of cardiac activity in everyday situations. Common artifacts due to movement or changes in illumination can surpass the signal by orders of magnitude and pose the main problem that hinders rPPG. Several signal processing approaches to improve the quality of rPPG signals were therefore proposed in the past [2, 3].

Another aspect of the signal extraction, the selection of the region of interest (ROI), has received little attention. In recent publications most commonly the entire face or static areas within are being used as ROI. Poh et al. determine

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the ROI using face detection. Hereby the rectangle enclosing the detected face is shrunk and then used as ROI [2]. Sun et al., Lewandowska et al. and Takano et al. use rectangular ROIs centered on the forehead and the cheeks [4, 3, 5]. While in the latter cases the selection was done manually, Lewandowska et al. also propose a method to geometrically determine a ROI on the forehead in relation to the position of the eyes. Verkruysse as well as Hülsbusch and Blažek have shown that not all parts of the skin's surface undergo the same amount of color change [6, 7]. Even though these phenomena have been described qualitatively, no quantitative analyses have been conducted yet.

2 Materials and methods

We hypothesize that it is possible to improve the rPPG signal's quality by determining a well-suited ROI. Building upon the aforementioned studies by Hülsbusch, Blažek and Verkruysse we investigate whether it is possible to define an optimized ROI using Lewandowska's approach of defining landmarks in the face. To achieve this we apply Verkruysse's method of pulse amplitude mapping [6], extending it by using precise reference signals.

2.1 Measurement setup

We conducted a study on 20 healthy subjects (age: 26.5 ± 4 years, sex: 8 female, 12 male). The subjects were asked to lie down on a tilt table and to hold still during the measurement. The camera (IDS uEye 5240 CP-C) was mounted 1 m above the table's surface, in front of the face. The videos were recorded synchronously with data of conventional, contacting medical sensors. We recorded RGB videos of 3 minutes length with a color depth of 10 Bit per channel, a resolution of 300 × 200 px and a frame rate of 100 Hz. The subject was illuminated frontally by four fluorescent tubes (NARVA LT 58 W/025 universalwhite). The distance between the lamps and the subject's face was 1.5 m. The lamps were equipped with electronic ballast circuits that drive the tubes with a frequency of 40 kHz avoiding the flickering with the line frequency. As ground truth reference

Fig. 1. Pulse amplitude maps showing the distribution of the rPPG signal's quality. Blue stands for a low value of the signal quality, red for a high value.